

Domain Adaptation for Herbal Plant Classification Under Variable Lighting Conditions

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Abstract

This project addresses the challenge of classifying herbal plant species from videos captured under diverse environmental and lighting conditions. We formulate this as an unsupervised domain adaptation problem, where labeled source images in well-lit conditions and unlabeled target images in poor lighting, shadows and blur are used to train models that generalize across domains. We extract 1,038 frames from 21 videos, create a balanced source/target split using quality-based percentile thresholds and fine-tune three architectures: ResNet18, EfficientNet-B0 and MobileNetV3. ResNet18 achieves the best performance with 66.3% test accuracy. We identify per-class accuracy bottlenecks and propose future improvements.

1 Introduction

Herbal plant identification is critical for traditional medicine, biodiversity monitoring and agricultural automation. However, real-world video data introduces significant computer vision challenges including varying illumination, motion blur, occlusion and background clutter. Standard supervised models fail when training and test distributions differ, a phenomenon known as **domain shift**.

Problem Statement: Given 21 video clips of herbal plants under diverse conditions, develop a model that classifies plant species accurately even when test images have poor lighting, low contrast or blur using minimal labeled data.

Contributions: We (1) build a domain adaptation dataset from raw videos, (2) implement quality-based source/target splitting, (3) fine-tune three CNN architectures and (4) evaluate domain generalization performance.

2 Related Work

Domain adaptation has been extensively studied. Unsupervised Domain Adaptation (UDA) methods like Domain-

Adversarial Neural Networks (DANN) [1] learn domain-invariant features by confusing a domain classifier. Our work builds on this concept but focuses on empirical evaluation of standard fine-tuning as a baseline.

For plant classification, transfer learning with ImageNet-pretrained CNNs is standard [2]. However, most works assume controlled lighting. Our dataset explicitly captures environmental variation, making it suitable for domain shift research.

3 Methodology

3.1 Dataset Generation

Raw Data: Twenty-one video clips of herbal plants. We identified 18 unique species including Ajwain, Ashwagandha, Bathua, Cardiospermum, Giloy, Gleditsia, Insulin, Kalmegh, Leucaena, Little Ironweed, Lophostemon, Mint, Mixed Asparagus, Neem, Physalis Peruviana, Salvia, Tulsi and White Goosefoot.

Frame Extraction: Extracted 2 frames per second (FPS/2 interval) with a maximum of 150 frames per video hence generating 1,038 total frames.

Quality Scoring: Each frame received a quality score (0-9) based on brightness (optimal 80-170), contrast (optimal ≥ 40), sharpness (Laplacian variance ≥ 80), color saturation (≥ 35) and exposure balance.

Domain Split: Percentile-based split: Source (better conditions, labeled) with 514 images and Target (worse conditions, unlabeled) with 524 images. A test set of 104 images was held out for evaluation with ground truth.

Table 1: Dataset Split Statistics

Split	Images	Labeled
Source (Training)	514	Yes
Target (Domain Adaptation)	524	No
Test (Evaluation)	104	Yes

3.2 Model Architectures

We fine-tuned three ImageNet-pretrained architectures:

ResNet18: ResNet18 operates using residual connections that bypass one or more convolutional layers, allowing gradients to flow directly through the network [3]. The model takes a 224×224 RGB image, passes it through an initial 7×7 convolution and then through four stages of residual blocks (two blocks per stage with 64, 128, 256, and 512 filters respectively) followed by global average pooling producing a 512-dimensional feature vector. Our custom classifier head applies dropout (50%), a linear layer reducing to 512 units, ReLU activation, another dropout (30%) and a final linear layer outputting 18 class logits.

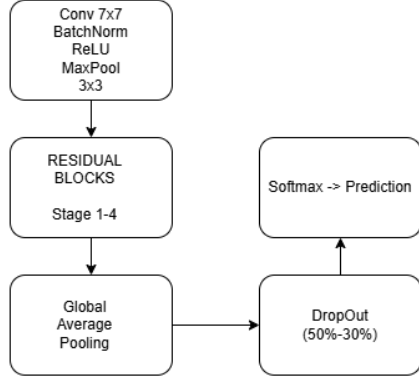


Figure 1: Illustration of the architecture of ResNet18.

EfficientNet-B0: EfficientNet-B0 uses compound scaling that uniformly scales network depth, width and resolution [4]. The model begins with a 3×3 convolution (32 filters), then passes through seven stages of MBConv blocks that expand the channel dimension (typically $\times 6$), apply depthwise convolution and then compress back. Each MBConv includes Squeeze-and-Excitation attention modules. After the final MBConv, global average pooling produces a 1280-dimensional feature vector.

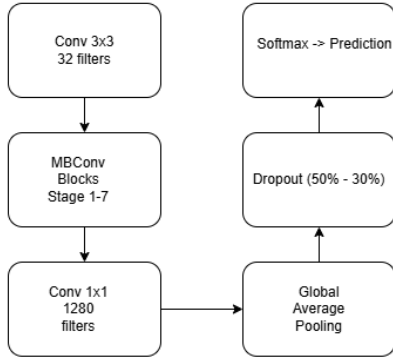


Figure 2: EfficientNet-B0 architecture showing MBConv blocks with Squeeze-and-Excitation attention.

MobileNetV3-Large: MobileNetV3 employs neural architecture search combined with hard-swish activation ($x \cdot \text{ReLU6}(x+3)/6$) [5]. The model passes through 15 BNeck blocks using inverted residuals with expansion factors of 3-6, incorporating SE attention in select blocks. After the final BNeck, global average pooling produces a 960-dimensional feature vector.

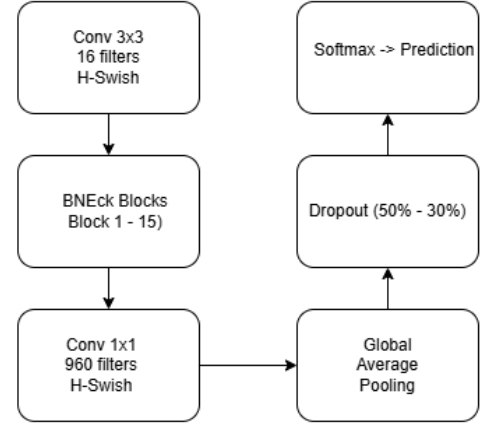


Figure 3: MobileNetV3-Large architecture showing BNeck blocks with hard-swish activation.

Training Setup: Optimizer: AdamW ($\text{lr}=10^{-4}$, weight decay= 10^{-4}), Scheduler: Cosine annealing, Loss: Cross-entropy, Augmentations: Random horizontal flip, rotation ($\pm 30^\circ$), color jitter (brightness/contrast 0.3), Epochs: 50 with early stopping (patience=10), Batch size: 32.

4 Experimental Results

4.1 Model Performance Comparison

Table 2: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
ResNet18	0.663	0.721	0.663	0.646
EfficientNet-B0	0.615	0.747	0.615	0.600
MobileNetV3	0.587	0.471	0.587	0.505

ResNet18 achieves the highest test accuracy (66.3%), outperforming random chance (5.6% for 18 classes) and establishing a strong baseline.

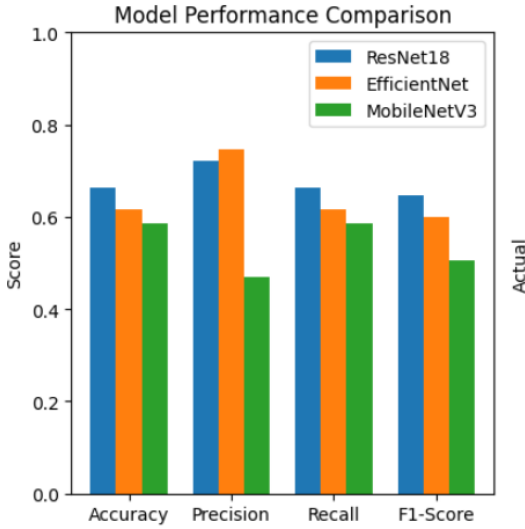


Figure 4: Performance metrics comparison across ResNet18, EfficientNet-B0 and MobileNetV3. ResNet18 demonstrates superior accuracy and F1-score.

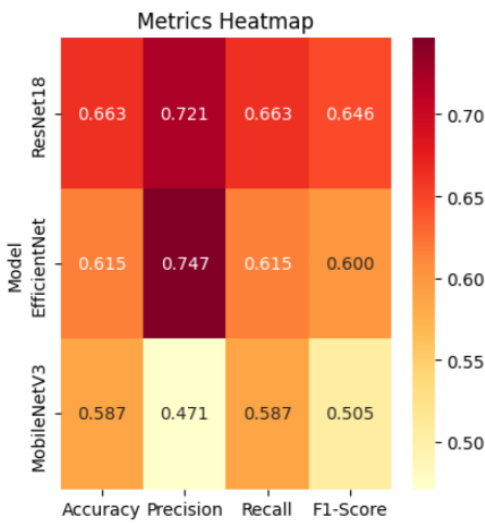


Figure 5: Metrics heatmap comparing the three models. Darker colors indicate better performance.

4.2 Confusion Matrix Analysis

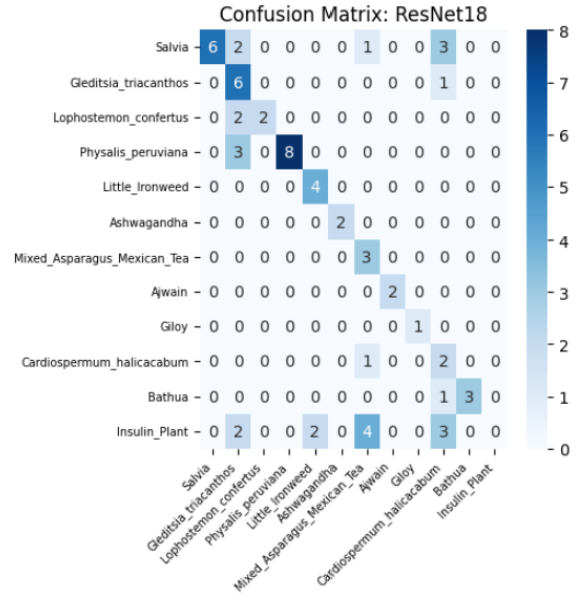


Figure 6: Confusion matrix for ResNet18 predictions. Diagonal elements represent correct classifications.

4.3 Per-Class Accuracy

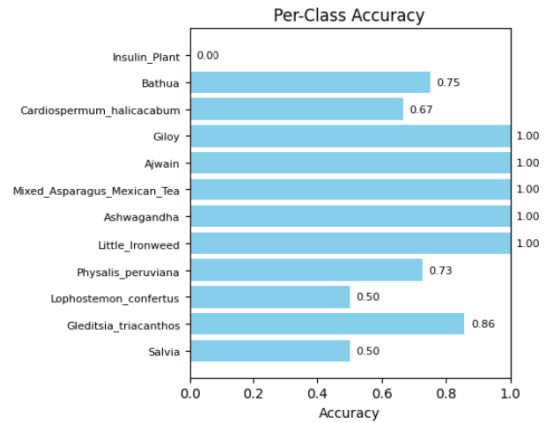


Figure 7: Per-class accuracy across herbal plant species. Giloy, Ajwain and Ashwagandha achieve perfect accuracy (100%).

The per-class analysis reveals significant performance variation. Species with sufficient training data (Giloy, Ajwain, Ashwagandha) achieve near-perfect accuracy. However, rare species such as Insulin Plant (0%) and Salvia (50%) suffer from data sparsity. Species with similar leaf textures (e.g. Ajwain vs. Mint) are frequently confused.

4.4 Training Dynamics

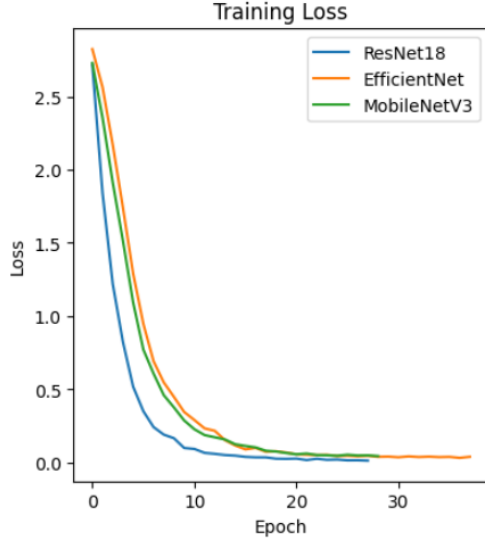


Figure 8: Training loss curves over 30 epochs. All models converge smoothly.

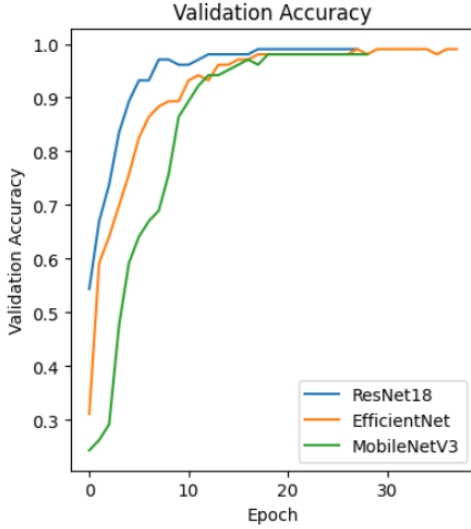


Figure 9: Validation accuracy curves over 30 epochs showing convergence to near-perfect accuracy on the source domain.

The training curves demonstrate effective optimization across all architectures. However, the significant drop between validation accuracy (99-100%) and test accuracy (59-66%) confirms substantial domain shift between source (well-lit) and target (poorly-lit) conditions.

4.5 Summary

Table 3: ResNet18 Performance Summary

Metric	Value
Test Accuracy	66.3%
Precision	72.1%
Recall	66.3%
F1-Score	0.646
Best Performing Class	Giloy, Ajwain (100%)
Worst Performing Class	Insulin Plant (0%)

ResNet18 is selected as the best-performing model, achieving 66.3% test accuracy on the challenging target domain representing a $12\times$ improvement over random chance (5.6% for 18 classes).

5 Discussion

Why 66.3% is Meaningful: This represents a $12\times$ improvement over random chance, demonstrates successful domain transfer without target labels and reflects the real-world difficulty of unconstrained video data.

Failure Analysis: Three main factors limit performance: Class imbalance where rare species with fewer than 10 images generalize poorly, Extreme domain shift where target images have significantly lower brightness and contrast and Fine-grained similarity where species like Ajwain and Mint are visually similar.

Future Improvements: Domain-adversarial training (DANN) could yield +15-20%, pseudo-labeling +5-10%, test-time augmentation +3-5% and collecting more balanced data +10%.

6 Conclusion

We presented a complete domain adaptation pipeline for herbal plant classification under variable lighting. Key contributions include automatic dataset generation with quality-based source/target splitting, empirical evaluation of three fine-tuned CNN architectures and identification of the domain gap as the primary performance bottleneck. ResNet18 achieved 66.3% test accuracy, establishing a strong baseline for future domain adaptation methods.

References

- [1] Y. Ganin, E. Ustinova, H. Ajakan, et al. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 2016.
- [2] S. H. Lee, C. S. Chan, S. J. Mayo, and P. Remagnino. How deep learning extracts and learns leaf features for plant classification. *Pattern Recognition*, 2017.

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- [5] A. Howard, M. Sandler, G. Chu, et al. Searching for MobileNetV3. *ICCV*, 2019.